



IIT PALAKKAD

COMPUTER SCIENCE AND ENGINEERING
Indian Institute of Technology, Palakkad

CS5658: GPU Programming

Aug - Dec 2025

Mid Semester Exam(25/09/2025)

Time: 75 minutes

General Instructions

- Please enter your name and roll number in the question paper.
- Maximum marks 50

Name _____ Rollno 112201008

1. ✓ The CUDA programming model extends C programming language. List any three keywords available in CUDA programming language and not in C, and used in the declaration statement of CUDA programming language. Give the description of these keywords. [6]
 2. ✓ Mention three variables specific to CUDA programming model and their meaning. [4]
 3. ✓ Write a CUDA kernel that implements `atomicAdd()` using `atomicCAS()`. Explain your code. [10]
 4. (a) Explain the terms: (i) ✓ sequential consistency, (ii) ✓ inter-iteration dependency (iii) ✓ intra-iteration dependency. (3 Marks)
 - (b) ✓ Transform the below code to get maximum parallelism. Justify your answer. (7 Marks)
- ```

for(int i=1; i<10; i++){
 X[i-1]=B[i]+i;
 D[i]=X[i]+i;
}

```
5. (a) ✓ What is the difference between thread scheduling on a multi-core CPU and a Nvidia GPU?. Explain (3 Marks) [10]
  - (b) ✓ Given implementation of same optimized Bellman Ford SSSP algorithm for : multi-core CPU (SSSP\_CPU) and Nvidia GPU (SSSP\_GPU). Is it possible that SSSP\_CPU takes less time for an input graph  $G_1$  and SSSP\_GPU takes less time for another input graph  $G_2$ ?. Both  $G_1$  and  $G_2$  has  $|V|=1024 \times 1024$  and  $|E|=1024 \times 1024 \times 4$ . Justify your answer with a pseudo code. (7 Marks) [10]
  6. ✓ Write *pseudo-code* for tiled matrix multiplication. Explain the code. Justify how the code benefits in improving the performance. Adding loop unrolling benefits in improving the performance of tiled matrix multiplication code? [15]
- Mention CPU or GPU.